

Part 2 of 2

Android Application: Parallel RLC Circuit

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In Part 1 of this article, we discussed a series RLC circuit. In this concluding part we discuss a parallel RLC circuit.

RLC ABOUT

SERIES PARALLEL READ

Parallel RLC circuit

AC

R L C

Supply voltage: 50

Resistor: 50

Capacitor: 100e-6

Inductor: 20e-3

frequency: 50

calculate

Fig. 21: Parallel RLC circuit example input

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Inductive Reactance: 6.28319 ohm

Capacitive Reactance: 31.83099 ohm

Impedance: 7.73424 ohm

supply current: 6.46476 A

IR: 1.00000 A

IL: 7.95775 A

IC: 1.57080 A

PA: 81.10149 Deg

Admittance Y: 0.12930 s

Conductance G: 0.02000 s

Inductive susceptance BL: 0.15915 s

Capacitive susceptance BC: 0.03142 s

Resonance Condition

Resonant Freq: 112.53954 Hz

current at resonance: 1.00000 A

Inductive Susceptance BL: -0.07071 s

Capacitive Susceptance BC: 0.07071 s

Fig. 22: Parallel RLC circuit example output

The parallel RLC circuit is exactly opposite to the series circuit. Here, the resistor, inductor and capacitor

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Band Width: 31.83099 Hz

fL: 97.74387 Hz

fH: 129.57486 Hz

Graphical output

Scale difference: 1 Hz

X-axis Range(simulated frequency range)

start Xvalue: 1.0 Hz

end Xvalue: 225 Hz

Graph customization:

Adjust axis labels: 15

BL BC B Y PCC PA

BW IL IC

Fig. 23: Parallel circuit resonance output with frequency simulation range between 1.0Hz and 225Hz

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current at resonance: 1.00000 A

Inductive Susceptance BL: -0.07071 s

Capacitive Susceptance BC: 0.07071 s

IL: 3.536 A IC: 3.536 A

Admittance: 0.02000 s

Quality factor: 3.53553

Band Width: 31.83099 Hz

fL: 97.74387 Hz

fH: 129.57486 Hz

Graphical output

Scale difference: 1 Hz

X-axis Range(simulated frequency range)

start Xvalue: 90 Hz

end Xvalue: 140 Hz

Graph customization:

Adjust axis labels: 15

BL BC B Y PCC PA

BW IL IC

Fig. 24: Default axis label as 15



are connected in parallel. Current and voltage are in-phase in the resistor; current lags voltage by 90° in inductor and leads by 90° in capacitor. Screenshot of the parallel RLC circuit example in an Android app is shown in Fig. 21. The example output of par-

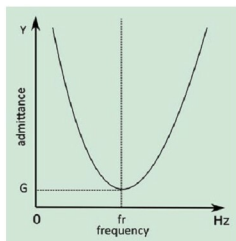


Fig. 25: Admittance versus frequency

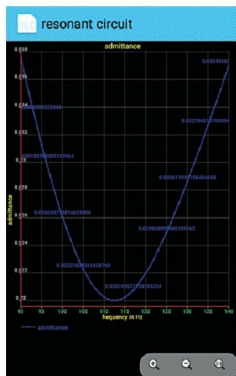


Fig. 26: Admittance versus frequency plot in Android app